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A wrist worn Internet-of-Things sensor node for wearable equivalent daylight illuminance monitoring

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***Abstract*—Light exposure is a vital regulator of physiology and behaviour in humans. However, monitoring of light exposure is not included in current wearable Internet-of-Things (IoT) devices, and only recently have international standards defined α*-opic equivalent daylight illuminance* measures for how the eye responds to light. This paper reports a wearable light sensor node that can be incorporated into the IoT to provide monitoring of equivalent daylight illuminance exposure in real-world settings. We present the system design, electronic performance testing, and accuracy of equivalent daylight illuminance measurements when compared to a calibrated spectral source. This includes consideration of the directional response of the sensor, and a comparison of performance when placed on different parts of the body, and a demonstration of practical use over 7 days. Our device operates for 3.5 days between charges, with a sampling period of 30 s. It has 10 channels of measurement, over the range 415–910 nm, balancing accuracy and cost considerations. Measured α-opic equivalent daylight illuminance results for 13 devices show a mean absolute error of less than 0.07 log lx, and a minimum between device correlation of 0.99. These findings demonstrate that accurate light sensing is feasible, including at wrist worn locations. We provide an experimental platform for use in future investigations in real-world light exposure monitoring and IoT based lighting control.**

***Index Terms*—Light sensing; Low power electronics; Wearable devices.**

I. Introduction

LIGHT is an external stimulus that the human body is exposed to on an everyday basis, and is a vital regulator of physiology and behaviour in humans [1]. This includes non-image forming responses in physiology and behaviour, such as regulation of the circadian clock, melatonin secretion, alertness, sleep, mood, and cognitive performance [2]–[5]. As a result, the amount of light exposure has acute non-visual effects, including on alertness, mood, and cognitive performance. A wearable light monitoring device is highly beneficial as it can provide light exposure data for each individual and for example, can potentially control lighting settings in Internet-of-Things (IoT) light enabled offices [6] to optimise light exposure for workers. However, monitoring of light exposure is not included in current wearable IoT devices. While wearables measuring activity levels and heart rate are an important part of wearable IoT devices, light sensing presents an important and unmet gap due to the challenges of adequately measuring light exposure in a way which replicates the response of the human eye with appropriate spectral weightings.

The human eye is capable of receiving visual images and converting them into electrical signals through two different groups of photoreceptors, namely, rods and cones. Rods are highly sensitive and function best at low light levels (scotopic), while cones are responsible for daytime and high-intensity (photopic) light levels and for colour vision. There are three types of cones, tuned for long, middle, and short wavelengths or, in short, L-cones, M-cones, and S-cones, and both rods and cones can be operational at intermediate (mesopic) light intensities, where rods gradually become sensitive while cones are still active, with overlapping spectral sensitivities [7], [8].

In addition, it is now known that the eye is also capable of detecting light independently of the pathways responsible for perception of visual images. This involves a subset of retinal ganglion cells that serve as photoreceptors called intrinsically Photoreceptive Retinal Ganglion Cells (ipRGCs), which were discovered in 2002 [9]. These cells express the light-sensitive retinal protein (opsin-like photopigment) melanopsin, which is predominantly sensitive to short-wavelength light with a peak sensitivity of $\sim$480 nm [10]. Convergent evidence from animal and human studies indicates that ipRGCs are the primary route by which light, acting via the circadian system or more acutely, can influence daily patterns of physiology, sleep, and alertness [2]–[5], [11], [12].

Measuring both types of light exposure (visual and non-visual) is therefore important to include in wearable and IoT

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light sensors. However, there has been uncertainty in how to best quantify non-visual responses since cone and/or rod photoreceptors may also contribute, alongside melanopsin, to non-visual responses under some circumstances. Several different metrics have been proposed to quantify light exposure input from different photoreceptors to non-visual systems with respect to these parameters [13]–[15], and lack of consensus on metrics combining the different forms of light response has delayed the incorporation of light sensing into wearable IoT devices. For example, in 2017 [16] reviewed commercially available light sensors, finding "*Optical performance was found to be quite varied, with each device having advantages and disadvantages*".

To resolve this, the International Commission on Illumination (CIE) has introduced standard CIE S026:2018 [17]. Here the effective irradiance for each of the five individual photoreceptors (rods, L-cones, M-cones, S-cones and ipRGCs) are referred to as the α-opic irradiance [18]. An α-opic irradiance can then be expressed as an Equivalent Daylight Illuminance (EDI, unit lx) corresponding to the illuminance of daylight required to produce that α-opic irradiance. Monitoring α-opic Equivalent Daylight Illuminance (EDI) thus provides a method for characterising light exposures for non-visual effects and is now the accepted and standard way of giving health and lighting recommendations [11], [12]. For example, varying lighting can influence sleep timing, melatonin secretion, and pupil diameter [19], while disturbances in circadian rhythms (heavily influenced by lighting) can lead to health issues such as sleep disorders, cardiovascular disease, and even cancer [20], [21]. Given this importance on human and animal health and wellbeing, a wide range of wearable light sensors (dosimeters) have been reported previously [22]–[26], and some commercial devices including light monitors, such as the LYS [27] are available. A comprehensive review is given in [28]. However, to date only a small number of works have attempted to make sensors informed by CIE S026:2018.

Light sensors that meet this new standard are now starting to emerge [29]–[33]. [29] used an AS7265X sensor (ams, Premstaetten, Austria) and 14 channels covering wavelengths from 410–760 nm to produce a light sensor in the form of a pendant round the neck. (Note the AS7265X contains 18 channels, only 14 of which were used.) The wearable had dimensions of $77\times46\times28$ mm, weighed 50 g, and with sampling every 30 s had a battery life of 23 hours for a real-world accuracy estimated as 17%. [30] developed a wearable light sensor designed to be worn as a pendant to measure Spectral Power Density (SPD) data in the visible wavelength range 380–780 nm using 256 channels. [31] developed a wearable light dosimeter using a light sensor with 6 channels covering a wavelength range of 380–780 nm, which can be attached to spectacle frames. This was updated in [32]. [33] described a sensor unit which produced CIE S026:2018 based measures in a miniature form factor, but focused on skin spectrum measurements (for example for the cosmetics industry) rather than as a light dosimeter for monitoring light exposure. Some commercial products are also starting to emerge, such as the XL-500 from nanolambda [34], and ActLumus from Condor Instruments [35]. As a result, [36] has recently introduced recommendations for the clinical validation of wearable light loggers.

This paper presents a new wearable IoT node for monitoring α-opic EDI and integration into IoT deployments. Unlike the previous CIE S026:2018 informed sensors above (except for [35]) it is designed to be worn at the wrist, in-line with most wearable devices. We present a new investigation into the different EDI estimations produced with the sensor placed on different parts of the body. We demonstrate that the wearable device is capable of accurately estimating photoreceptor-specific illuminations with a mean absolute error of less than 0.07 log units across the range of light intensities encountered between full moonlight and bright sunlight. Cost considerations are a major driver in our design approach, and a channel count optimization is considered in order to balance α-opic measurement accuracy with complexity and power consumption. Further, we demonstrate inter-and intra-device reliability. Collectively, the data demonstrate our system is suitable for large-scale longitudinal studies designed to assess how individuals' ambient light exposure profile impacts sleep, circadian rhythms, and alertness in their daily life. The current paper focuses on the system design and controlled performance testing results. [37] reports a 59 participant experiment making use of the new IoT technologies introduced here.

The remainder of this paper is organised as follows. Section II overviews the hardware design of the new IoT device. Section III then presents the analysis methods used for evaluating the performance of the device, and Section IV presents the measured performance, in terms of accuracy, reliability, and with placement location on the body. Finally, conclusions are drawn in Section V.

II. System design

A. Electronic hardware

An overview of the system architecture is given in Fig. 1. The wearable device is powered by a Lithium-ion rechargeable cell, with USB-C connection for charging via a custom charging board. A BLE652 (Laird connectivity, Akron, Ohio, USA) provides an Arm Cortex-M4 central processing unit and Bluetooth 5 connectivity. A microSD card slot is included for redundancy and data logging experiments where wireless streaming of data is not required. For auxiliary sensing, an ADXL362 accelerometer (Analog Devices, Wilmington, Massachusetts, USA), and a low-power, high-accuracy digital temperature sensor (MAX31889, Maxim, San Jose, California, USA) are used. These have accuracies of 1 mg/Least-Significant-Bit, over ±2 g, and $0.25^{\circ}\mathrm{C}$ over $-40^{\circ}\mathrm{C}$ to $+125^{\circ}\mathrm{C}$, respectively.

Light sensing is provided by an AS7341 multi-spectral spectrometer (ams, Premstaetten, Austria). It provides a high-precision optical filter on standard CMOS silicon, covering approximately the 350–1000 nm range with 8 channels centred in the visible spectrum, plus a near-infrared and a clear channel. Channel peak sensitivities are at 415 nm, 445 nm, 480 nm, 515 nm, 555 nm, 590 nm, 630 nm, and 680 nm, with Full Width Half Maximum (FWHM) values of 26 nm, 30 nm, 36 nm, 39 nm, 39 nm, 40 nm, 50 nm, and 52 nm

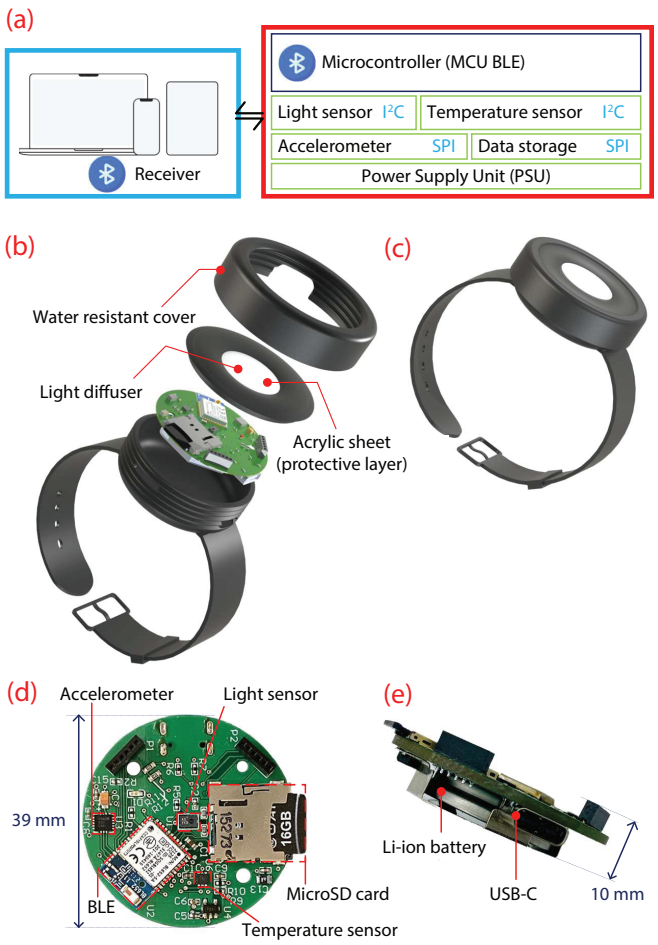


Fig. 1. The wearable light sensor consisting of a 10 channel spectral light sensor, temperature sensor, accelerometer, Microcontroller and Bluetooth low energy (MCU BLE) integrated module, lithium-ion battery, and microSD card assembled on a custom designed Printed Circuit Board (PCB). (a) System architecture. (b) Exploded view of the case, including showing the light diffuser layer. (c) Assembled view of the case. (d) Photograph of the fabricated electronics, top view. (e) Photograph of the fabricated electronics, side view.

respectively. This unit was selected to trade-off cost and accuracy, with software processing (discussed in Section II-C) used to calculate α-opic EDIs. Using prices at the time of writing, the AS7341 is available in low volumes for £9 per unit, compared to £162 for the nanolambda NSP32m used in [33] and [34]. Our low cost approach is intended to be compatible with large IoT deployments and crowd sourcing experiments, and we demonstrate in Section IV that acceptable sensing accuracy is still obtained.

The layout of our device was sized to match that of a commercial wearable device, with dimensions (diameter $\times$ height) of 39$\times$10 mm and weight of 15 g (excluding enclosure), and 47$\times$16 mm and weight of 40 g (including enclosure). In comparison, the Samsung Galaxy Watch 4 Classic comes in 42 and 46 mm sizes, with a weight of approximately 50 g.

B. Enclosure

Enclosure design is particularly important for a light sensing device, with the need to include holes for light penetration (and charging), and careful design of the light penetration aperture

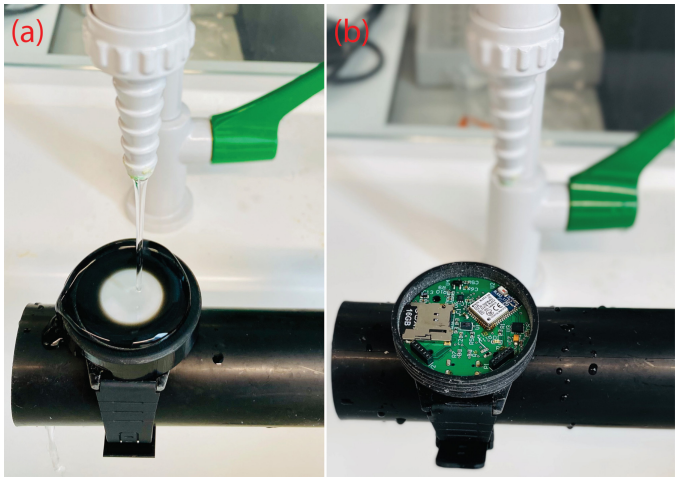


Fig. 2. Water resistance test of the developed wearable light monitoring device. (a) During (under the tap water flow). (b) After the water resistance test.

to set the light properties and range of angles of operation. This is combined with user needs and preferences, comfort and cost considerations, and the need for suitable water robustness during rainy weather and user activities such as hand washing.

To manage cost and allow user customisation, we opted for a 3D printing based approach. The case was made using a Stratasys J750 3D printer with Vero black resin, and a commercial fabric strap running behind the enclosure, sitting in between the skin and the enclosure. A screw top design as shown in Fig. 1b was used to allow easy access to the SD card and similar, with the cover blocking the USB charging port during use to prevent water ingress, and then being rotated by approximately 30 degrees to allow charging. Fig. 2 shows the water resistance test of the developed device. As shown, the enclosure preserves the device from the water penetration from the top. Nevertheless, users are advised to avoid getting the unit wet during use.

Light enters the device via a 2 cm cut-out in the inner cover (Fig. 1b). It first passes through an acrylic disc (Perspex) with diameter 30 mm and thickness 1 mm which acts as a protection layer. There is then a diffuser (Optsaver L-52, Kimoto, Cedartown, Georgia, USA) attached 2 mm above the light sensor. The acrylic disc is transparent to more than 92% of irradiance between 350–950 nm and the diffuser transparent to more than 68% of irradiance between 400–950 nm. A silicone sealant closes the gap between the diffuser/protective layer and the enclosure lid.

C. Software and sensor gain calculation

In its default configuration, the sampling period is set to 30 s, with each data acquisition requiring a maximum of 2 s (as discussed below). Data logging is started and stopped by a companion Android App (not shown here), which supports real-time data streaming of measured values over Bluetooth Low Energy. In the case that only data logging is required, the Bluetooth can be disconnected after the recording has been started, and the data is then logged to the SD card. Bluetooth Low Energy advertising continues during this data collection

to allow the App to re-connect. The advertising period is set to 2 s, allowing the device to be in a power saving mode for 93% of the time.

For light measurements, the sensor has two main controllable parameters: the integration time and gain. Natural daylight has a dynamic range in excess of 100 dB, such that it is not possible to cover all input light values with a single setting for integration time and gain without saturation. We fix the integration time at 0.182 s and use a search algorithm to find the best (maximum possible) gain between the maximum and minimum range supported by the AS7341. That is, the sensor's raw output is placed as close to the maximum as possible without saturation, which is highly desirable for measuring light in dark or dimmed places. For each measurement, first, a middle gain value is selected from the AS7341 gain range (8×, ..., 512×) and taken as the current gain. (The minimum was set to 8× although the hardware supports lower.) A prototype measurement is taken, and if the raw value is above the maximum range of the raw values (i.e. saturated), the gain is reduced by half. Otherwise, if the raw value is below the minimum range of raw values (the analogue-to-digital converter's noise limit), the gain correction is made by increasing the gain to the maximum and repeating the cycle. We allow up to 2 s to perform each sampling cycle, to allow all different gain settings to be trialled in order to optimise the dynamic range.

D. Equivalent Daylight Illuminance (EDI) calibration

The sensor's raw output is in *counts* and we converted this to α-opic EDIs defined in CIE S026:2018 [17] using offline processing in MATLAB (The Mathworks, Natick, Massachusetts, USA). This used a data driven approach, fitting calibration data collected from the wearable devices with a calibrated bench top light source and spectroradiometer (SpectroCAL MKII Spectroradiometer, Cambridge Research Systems, Cambridge, UK).

Calibration data was collected across 70 light conditions (5 distinct spectra, each at 14 different irradiances) provided by a custom built calibrated reference light source. The reference source (components from Thorlabs, Ely, UK) combined light from high power violet and white LEDs (with a selectable band pass filter) via a dichroic mirror to produce light in the following spectral ranges (centre wavelength ± half bandwidth): 403±9 nm ('violet'), 465±14 nm ('blue'), 550±20 nm ('green'), 602±11 nm ('amber') and 575±65 nm ('broadband yellow'). Produced light spectrums using the custom light source are given in the results in Section IV-B. LED intensity was controlled by a combination of LEDD1B T-cube drivers and a neutral density filter wheel to provide light intensities in the range $10^{10.5}$ to $10^{16.5}$ photons/cm^2/s, as determined via the calibrated spectroradiometer. All measurements were performed in a dark box. The devices were exposed to each light condition for 30 seconds and average sensor counts were calculated for each condition.

Prior to the use of each new wearable device we first measured α-opic EDIs using the calibrated light source and spectroradiometer, and counts using the new wearable. We then used nonlinear least-square fitting to derive weighting coefficients for the 10 channel count readings for each wearable that best recreated the α-opic EDIs from the calibrated spectral irradiances. Based on initial exploratory analysis, calibration data dimmer than 0.1 lx for every α-opic quantity was omitted, and fits were repeated across random subsets of the remaining data (36/52 measurements, 2500 iterations) to avoid convergence to local minima. Weighting coefficients from the overall best fit model for each device were then used to calculate estimated α-opic EDIs (with a floor at 0.1 lx).

III. Performance evaluation methods

A. Electronic performance

Power consumption was measured with a Keithley 2281S-20-6, precision DC supply and battery simulator in place of the Lithium ion battery. Voltage and current measurements were performed at 432 Hz with the device in always-on mode, and in typical usage mode with sampling every 30 s and the device put into sleep mode between samples.

B. EDI measurement accuracy

To test the accuracy of the AS7341 as the core sensing unit, we drove the sensor with 5 light spectrums using the custom light source described in Section II-D. These were designed to match the nominal peak wavelengths of each channel in the AS7341: Channel 1 (415 nm) 'violet'; Channels 2 and 3 (445 nm and 480 nm) 'blue'; Channels 4 and 5 (515 nm and 555 nm) 'green'; Channels 6 and 7 (590 nm and 630 nm) 'amber'; and Channels 8–10 (680 nm, clear, near infrared) 'broadband yellow'. We then compared the normalised log sensor counts with the normalised log optical power provided by the input source via a linear fit with outliers removed.

To test the validity of our calibration procedure for estimating α-opic EDIs from the sensor count values, we compared the measured EDIs from three wearable units and a calibrated spectroradiometer (SpectroCAL MKII Spectroradiometer, Cambridge Research Systems, Cambridge, UK) using data collected at 79 different physical locations with varying intensities of natural and/or electrical (LED, incandescent, RGB or fluorescent). Using a rod and clamp, all measurements were collected from the same physical locations for our sensor and the spectroradiometer, with locations reflecting diverse spectral light environments encountered in real-world settings. Expected and estimated log α-opic EDIs were compared using linear regression, with errors calculated as mean absolute log deviation.

C. Sensing optimization

We investigated a number of different factors in order to optimize the sensing carried out: the directional response, wear location, sampling period, and the number of channels used in the light sensor (default 10).

To establish the directional response properties, one of our devices was attached to a rotating board, 20 cm away from a white light source (XP-L white LED on star board,

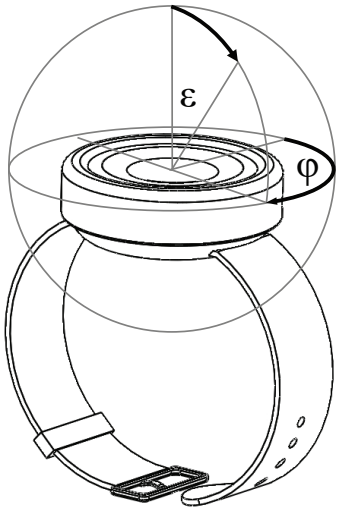


Fig. 3. Schematic representation of the co-ordinate system.

CREE LEDs, Opulent Americas). Using the co-ordinate system shown in Fig. 3, at angles ϵ between $-90°$ to $90°$ light sensor readings were collected and converted to melanopic illuminance. Half-width at half maximum was calculated using a cubic spline fit line.

To assess the impact of varying the device wearing position (wristband, pendant, forehead-attached) on the measurement outputs, three devices were simultaneously worn by a participant at the wrist, chest, and forehead. Five repeats of 2 minute measurements were collected from various indoor and outdoor locations for different body positions, activities and wrist directions. Our aim was to make a wrist worn wearable, acceptable by users in real-world conditions and mirroring smartwatches which are widely used and accepted in this wear location. Nevertheless, wear location may have a substantial impact on the measured light.

To minimise power consumption the number of light samples taken is ideally minimised, at the potential cost of precision. Our integration time of 182 ms means that high frequency components from ambient light sources at mains frequencies (50/60 Hz) and their harmonics cannot be measured using our approach. (Note that a flicker detection channel is available in the AS7341.) The maximum sampling rate supported by our software set up (Section II-C) is 0.5 Hz, once every 2 s, with this intended to allow the monitoring of long term changes in light exposure over the course of a day. However, the required Nyquist sampling rate for long term light exposure is not known *a priori*. Considering that ambient light does not usually change very rapidly in the real world we hypothesised that slower sampling rates would still yield acceptable measurements of light changes over time. To evaluate the impact of sampling rate, one device (sampling period 2 s) was worn for 200 hours (over 3 weeks) as a wristband by a participant during the daytime. The data was separated into 10 minute bins and mean, maximum, minimum and variance were calculated and compared for every bin using varying sampling periods (2 s, 10 s, 30 s, 60 s, 120 s, 300 s) by downsampling the 2 s raw data.

Finally, we repeated the above analysis, but rather than changing the sampling rate changed the number of light sensing channels used in the EDI fitting procedure. The used sensor has 10 channels, and we evaluated the impact of using a reduced number of channels for the daytime light exposure, uncontrolled over the 200 hour data collection period.

D. Real-world feasibility test

In a real-world feasibility study of the final version of the device, one participant wore the device for one week. The sampling period was set to 30 s (based upon the results from the analysis described above). This individual was asked to wear the device during daytime, to charge the battery every night, to pay attention not to cover the device with clothes sleeves, and to protect against water. Beyond this, the participant was asked to continue their regular daily activities. Sensor readings were converted to α-opic EDIs. All procedures were approved by the University of Manchester Research Ethics Committee (application number 2021-12948-20856) and participants gave written informed consent before taking part.

The current paper focuses on the system design and controlled performance testing results, and hence the scope covers only a single real-world feasibility test to demonstrate operation. [37] reports a 59 participant experiment making use of the new IoT technologies introduced here.

IV. Results

A. Electronic performance

Fig. 4 depicts the measured power consumption profiles of a typical device. The always-on mode is when the Bluetooth module is continuously ready to connect (advertising), while in typical usage the Bluetooth module only advertises for 2 s within each 30 s measurement period, with data logged on the SD card. As shown, the average power consumptions are 26.5 mW and 5.5 mW, respectively. Using a 120 mAh battery the device lasts for up to 3.5 days in typical usage.

B. EDI measurement accuracy

Fig. 5a shows the spectral distributions of the reference light source used as the input for the controlled EDI measurement tests, as determined via a calibrated spectroradiometer. Fig. 5b then shows how the output from each channel of the AS7341, in counts, varies as the input optical power from the light source is varied (outliers removed). As expected, Fig. 5b shows that all channels display a linear relationship between log normalised counts and the log optical power they were exposed to. All fits have an R^2 value $>$0.98. Fig. 5c shows that the distribution of normalised sensor counts were similar to the spectral distributions of the illuminants. Across our devices, there was a linear relationship between normalised sensor counts, before the conversion to EDI, and normalised optical power across $>$5 orders of magnitude and with slopes $\approx$ 1 (mean$\pm$Standard Deviation (SD): 1.02$\pm$0.04)

The accuracies of the estimated α-opic EDIs calculated by 13 different wearable devices are shown in Fig. 6. All estimated α-opic EDIs show linearity with the provided input

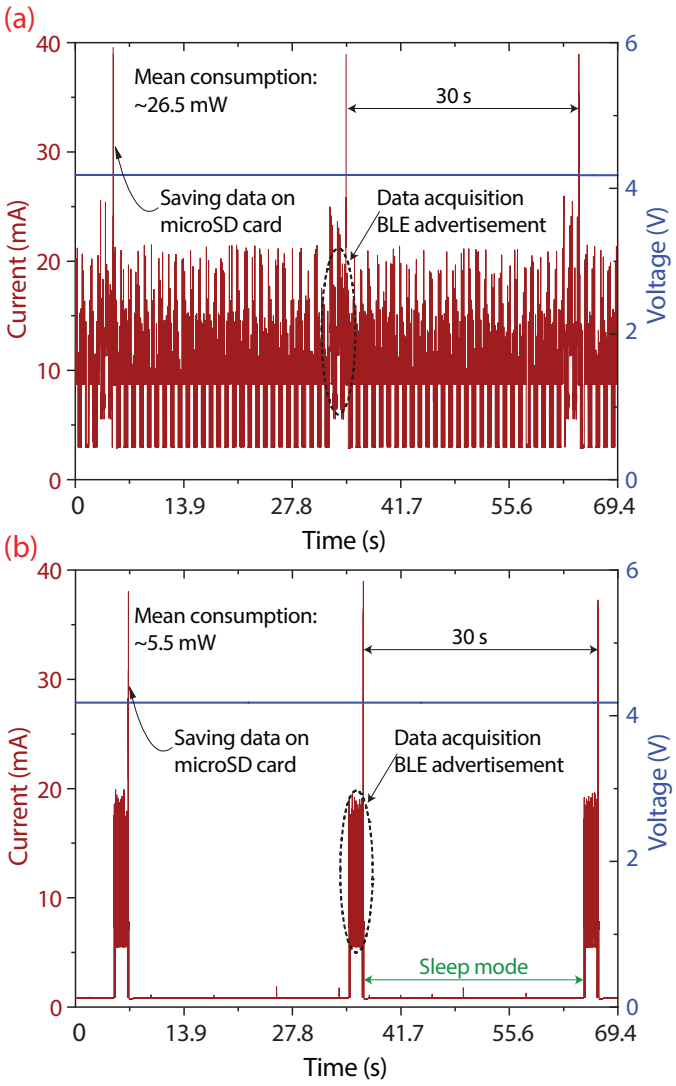


Fig. 4. The measured device power consumption profile for two measuring cycles with 30 s sampling period. (a) Always on mode with continuous BLE advertising. (b) Typical usage mode with reduced advertising rate and sleep mode used to reduce power consumption.

values. These are compared by linear regression, with in Fig. 6a R^2 values of 0.85, 0.98, 0.99, 0.98, 0.98 and 0.95 for S-cone-opic, Melanopic, Rodopic, M-cone-opic, L-cone-opic and Photoic EDIs respectively; and in Fig. 6b all R^2 values >0.99. Using errors calculated as mean absolute log deviation and all 10 channels in the regression model, for α-opic EDIs (or photopic illuminance) above 1 lx there was an average Mean Absolute Error (MAE) <0.07 log units and a minimum between-device correlation of 0.99. There were minor variations in performance across different α-opic quantities, with slightly reduced accuracy for S-cone-opic and greater accuracy for melanopic EDI (median±SD MAEs: 0.096±0.027 and 0.054±0.016 log units respectively, $n = 13$ devices). The typical performance of the device allowed us to reconstruct reliable estimates (within ±15%) of α-opic EDIs between 1 and >50,000 lx, a range encapsulating all practically relevant light exposure likely to be encountered during everyday life. Moreover, the estimation algorithm had a high within-device reliability. Repeated measures collected using the same device resulted in highly correlated outcomes (within device MAE being 0.03, range: 0.01–0.06).

The real-world accuracy, comparing measurements from three wearable devices to a non-wearable calibrated spectroradiometer is shown in Fig. 6b. Our algorithm and devices successfully estimated α-opic EDIs in real-world conditions (5–60,000 photopic lx) with LED, incandescent, RGB, fluorescent and daylight illuminants. All field measurements together had a mean MAE 0.08 (range: 0.05–0.14) and near perfect linear relationship with the expected values (mean slope of 1.03). We did not have enough sample size in fluorescent and incandescent illuminants, but estimations under LED, RGB, natural and mixed lights were similarly accurate.

C. Sensing optimization

As expected for the optical design of the devices and integrated diffuser, the angular response to incident light was near-cosinusoidal with a half-width at half maximum of 51° (Fig. 7a). This wide angular responsiveness enables us to use the device by attaching it to different body parts. Nevertheless, it is impossible to fully capture ocular light exposure.

Measured illuminances were influenced by the wearing position of the sensor as shown in Fig. 7b. Wrist-based measurements were generally lower than those obtained from a head mounted sensor, reflecting the measurand being different for the different sensor locations. Mean deviation from the forehead measurements were compared using a one-way ANOVA with Dunnett correction for multiple testing ($F(2,2877) = 18.39$, p <0.001). Wristband observations were significantly lower compared to forehead measurements (*** p <0.001), while wearing the device in a pendant location around the neck provided similar mean measurements with the forehead location (ns, non-significant). However, the magnitude and direction of the effects were context dependent, as shown in Fig. 7c. Wrist-based measurements often overestimated ocular illuminance when seated indoors with the device facing upwards, while they underestimated it, either indoors or outdoors, when the device was facing downwards or to the side. However, we should note that these observations are valid for highly stationary test cases, which do not usually not occur in daily life.

Our results from 200 hours of continuous wear showed that sampling intervals between 10 s and 60 s resulted in similar mean α-opic EDIs compared to 2 s measurements within any given analysis window (regression slopes >0.99). However, the ability to capture variance in the lighting environment decreased as the sampling interval period increased, with only 10 s and 30 s sampling intervals giving similar estimates of variance compared to that obtained for 2 s sampling (regression slopes >0.90). Increasing the sampling period also lowered the accuracy with which the minimum and, to a lesser extent, maximum α-opic EDI within a given sampling epoch could be ascertained. Based on this analysis we concluded that 30 s sampling provided a good compromise to maximize battery life for the light monitors without discarding substantial information about finer grained temporal changes.

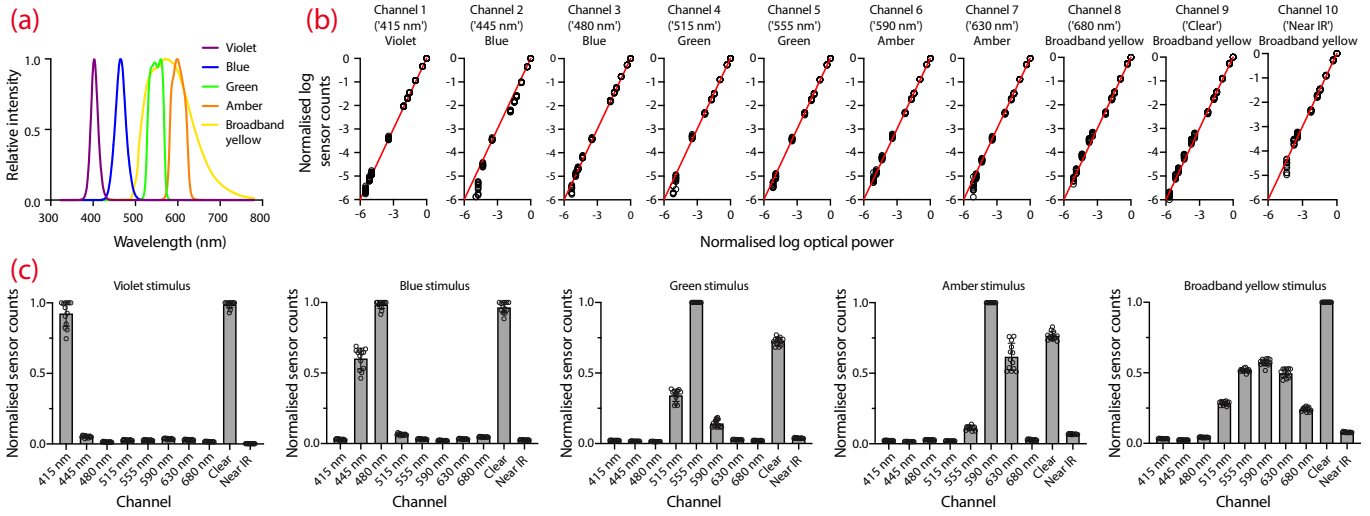


Fig. 5. Sensor measurement accuracy. (a) Spectral distributions of the reference input light source as determined via a calibrated spectroradiometer. (b) Comparison of normalised log sensor counts by each channel in the sensor vs. normalized log input optical power, when each channel is driven by its optimal reference illuminant match. Outliers removed. The red lines show the expected linear relationship (slope = 1). (c) Distribution of normalised counts across 10 channels for all devices. Sensor counts are normalised according to the maximum sensor reading of each device. The dots show each normalised sensor count with overlapping points horizontally offset, the shaded bars show mean, and the error bars show standard deviation.

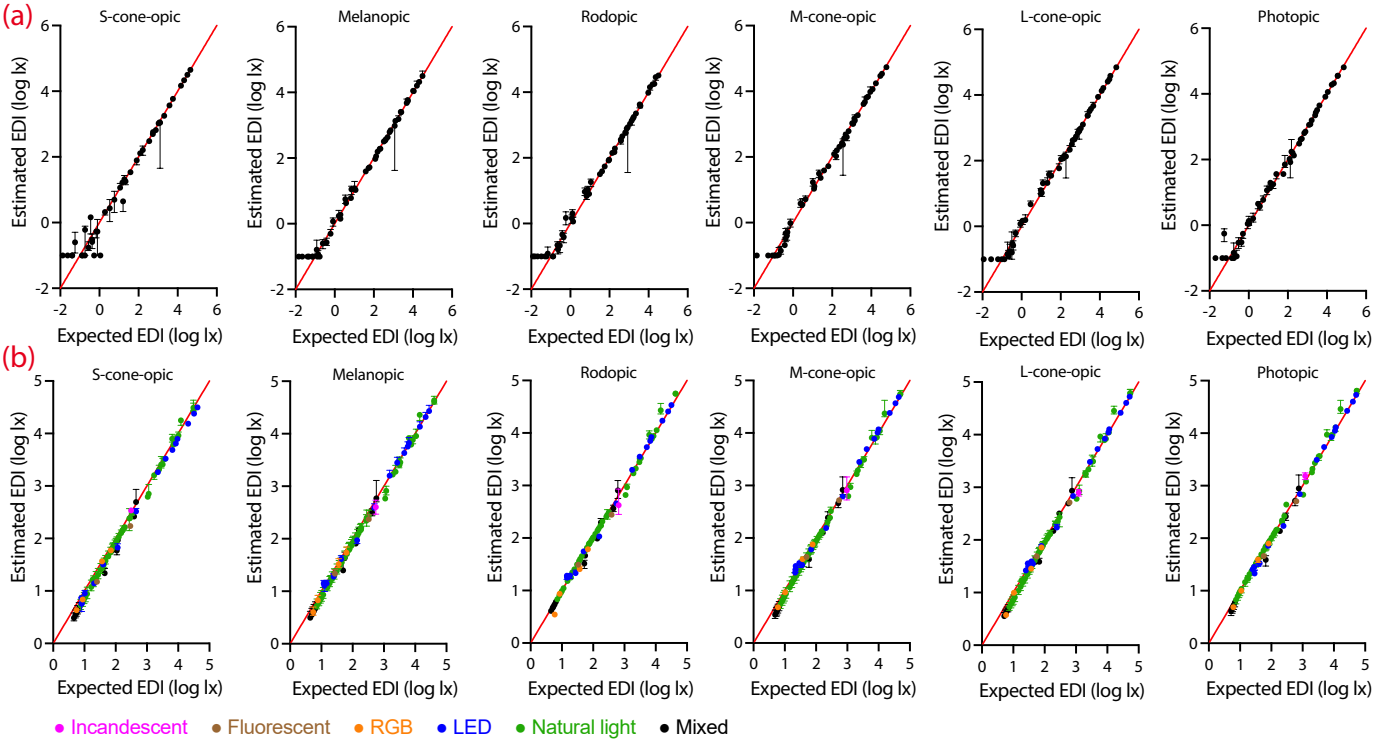


Fig. 6. EDI estimation accuracy. (a) Expected and estimated log α-opic EDIs of 13 devices (using the calibration set up) compared to each other to check between-device reliability and error rate. The dots show estimated log α-opic EDIs of 13 devices ($\pm$range). The red lines show the expected linear relationship (slope = 1). (b) Validity of three devices in real-world settings (vs. measurements by a calibrated spectroradiometer). The dots show estimated log α-opic EDIs of 3 devices ($\pm$range). The red lines show the expected linear relationship (slope = 1).

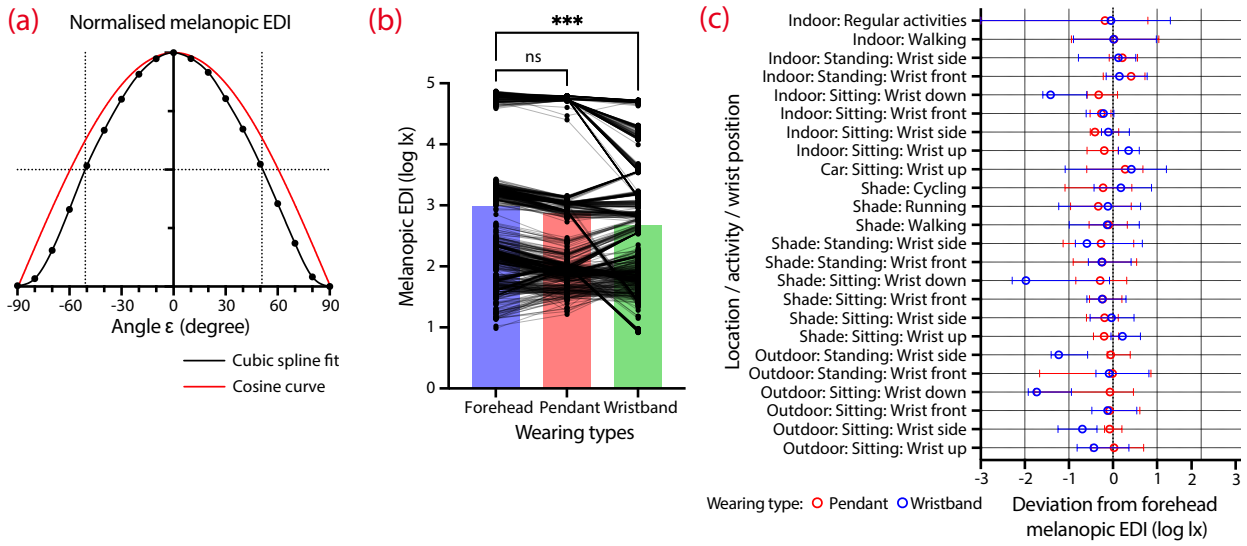


Fig. 7. Sensing optimization results. (a) Directional response of our device. The dots show normalised melanopic EDI of one device at angles ϵ between $-90°$ to $90°$ in front of the light source. The dotted horizontal black line shows the mid-point of the normalised melanopic EDI, and the vertical dotted lines show half-width at half maximum of a cubic spline fit line. The red line shows an expected cosine curve. (b) Wear location (head vs. neck vs. wrist) comparison. Five repeats of 2 minute measurements collected from 24 indoor and outdoor locations for different body positions, activities and wrist directions. The coloured bars show the average of all measurements, and the dots show all observations. The simultaneous measurements from different wear locations are connected with black lines. *** p <0.001, ns non-significant. (c) Deviation of neck and wrist measurements from head measurements in different locations/positions. The dots show median deviation $\pm$ range.

As stated in Section IV-B, using all ten channels in the AS7341 in the EDI calculation gave an average MAE <0.07 log units and a minimum between-device correlation of 0.99. Analysis of device performance based on subsets of the ten channels available indicated that performance dropped off exponentially as the number of distinct active channels was reduced. Reasonable accuracy could be obtained using only the nine narrowband sensors (i.e. excluding the clear channel), but this still resulted in an increase in MAE compared to the full models, especially in L-cone-opic EDI estimation (MAE 0.09 vs. 0.06 log units respectively for this band). We, therefore, used the full ten channel input model in all subsequent analyses. Overall, the measured dynamic range was 0.1—120,000 lx, covering the range from 0.1–0.3 lx of moonlight [38] to the 100,000 lx of direct sunlight (with 1 Sun typically being taken as 120,000 lx) [39].

D. Real-world feasibility test

The device collected 20,356 observations of light (169.6 hours, or 7.1 days) with an example day of wear shown in Fig. 8. The average accelerometer magnitude vector was 0.6 m/s^2 (SD = 1.0), see Fig. 8a. Total non-wear duration was 70.1 hours, consisting of a minimum 30.2% and maximum 52.7% of daily wear times. Total daily duration spent in dim conditions (median <1 photopic lx in 30 min) was a minimum of 6.7 hours, and maximum of 11.9 hours. Using the accelerometer we checked whether the device was placed face down and therefore getting a limited amount of light due to being occluded (median z-angle below $-60°$ and median <10 photopic lx in 5 min) and such instances were not recorded. Our observations confirmed that occlusion of a wrist placed sensor by clothing sleeves was a risk that would impact measurements, but this was also true for the pendant version (e.g. if users cover it by donning additional clothing) with, subjectively, the pendant location less comfortable for routine use in daily life. The ambient temperature sensor recorded an average of 20.1°C (range 7.0–28.9°C), Fig. 8b.

The device longitudinally and continuously measured α-opic EDIs, as shown in Fig. 8c for the melanopic range. Using this example as a demonstration of feasibility, we can extract several light exposure characteristics of the participant. Using weekly data, non-parametric variables (such as inter-daily stability, intra-daily variability or relative amplitude) of light exposure may be calculated [40]. The brightest 10 hours of melanopic EDI (known as M10) had 1.7 log lx with a mid-point between 16:00–17:00. The dimmest 5 hours of melanopic EDI (L5) had -1 log lx with a mid-point between 03:00–04:00.

Using weekly full data, descriptive statistics may also be calculated. The median melanopic EDI was measured as 6.4 lx (range: 0.1–123,083.1 lx). The participant was exposed to lower melanopic EDI compared to photopic illuminance (ratio: 0.8). This is expected considering that usual artificial illuminants have lower melanopic components compared to photopic illuminance, and indeed Fig. 8d shows the ratio decreasing in evening hours. In addition, weekly bright melanopic EDI (>1000 lx) exposure per day was 93.7 minutes. This was higher in weekend days (194.0 minutes) than in workdays (43.6 minutes). Moreover, daily harmonic fit model outputs can be calculated [40]. For example, on the day shown in Fig. 8c) the mesor of melanopic EDI trigonometric fit line was 0.7 log lx with an amplitude 1.0 and acrophase at 15:19. This day had a median melanopic EDI 10.3 lx (max 11,312.5 lx). The morning (4:00–12:00) median melanopic EDI was 7.7 lx (max 289.1 lx).

TABLE I
COMPARISON OF ACADEMIC WEARABLE DEVICES FOR MEASURING α-OPIC EQUIVALENT DAYLIGHT ILLUMINANCES (EDIS) INFORMED BY CIE S026:2018.

Ref.	Body placement	Light channels	Wavelength range (nm)	Sampling period (s)	Headline error	Battery capacity (mAh)	Battery lifetime (h)	Weight (g)
[29]	Necklace pendant	14	410–760	30	17%	—	23	50
[30]	Clip pendant	256	380–780	60	7%	110	24	20
[31], [32]	Mounting to spectacle frames	6	380–780	10	—	230	168	27
This work	Wrist	10	415–910	30	15%	120	84	40

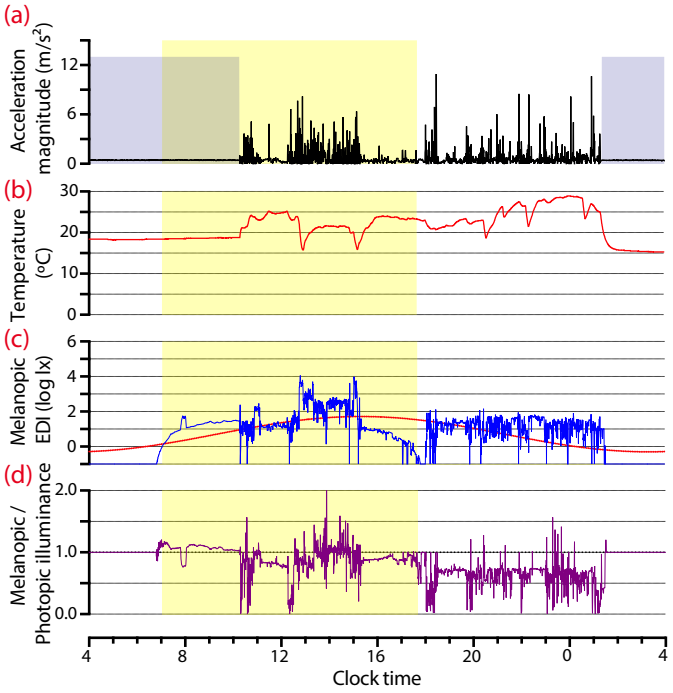


Fig. 8. Example of one day of recording activity, temperature, and light exposure. (a) Acceleration magnitude ($\sqrt{x^2+y^2+z^2}-9.81$) where x, y, z are the three accelerometer axes. The yellow area shows daylight duration (from sun rise to sunset). The blue regions show estimated non-wear time (defined as accelerometer average magnitude standard deviation $<$0.05 m/s^2 and acceleration vectors (x, y, y) standard deviation $<$5 mg). (b) Temperature recording. (c) Estimated melanopic EDI. The blue line shows melanopic EDI (log lx) throughout the day. The red line shows a daily cosine fit curve ($R^2 = 0.38$). (d) The ratio of melanopic EDI (lx) and photopic illuminance (lx).

The device successfully captured sun rise at around 7:00, which may indicate the participant had a window in the room but with a limited access to daylight since the measured $\sim$25 lx melanopic EDI around 10:00 (before waking up) may be lower than suggested morning light exposure [12]. The participant experienced bright light exposure, most likely outdoor activity, with more than 1000 lx melanopic EDI from 13:00 to 15:00. The evening (20:00–4:00) median melanopic EDI was 7.0 lx (max 144.2 lx). The participant experienced dusk around 18:00, however, from 18:00 to 2:00, unhealthy (late and bright) artificial light exposures (10–100 lx melanopic EDI) occurred [12]. Note that for compactness only melanopic EDIs are shown in Fig. 8. Outputs were not limited to photopic and melanopic illuminances, with all α-opic EDIs and activity continuously measured without interruptions.

V. DISCUSSION

Table I illustrates a comparison of state-of-the-art wearable devices in the academic literature for measuring α-opic equivalent daylight illuminances (EDIs) according to CIE S026:2018, with a wider review of light dosimeters (including those not monitoring according to CIE S026:2018) given in [28]. Table I indicates that clear trade-offs are present in the design and implementation choices made.

Firstly, each device in Table I uses a different approach for the most suitable place to wear the device. Although in principle the best measurement point is in the near-corneal plane as it is the best available proxy to retinal irradiation, other parameters like social acceptability, device dimensions, and comfort also needed to be considered. Ours is optimized for the wrist to match standard smartwatch locations. This paper has quantified the differences obtained to devices placed on other parts of the body for the first time (Fig. 7). Secondly, while each device in Table I has a similar lower wavelength of response, ours goes up to 910 nm to provide monitoring of near infra-red light and use of this in the EDI calculations. Thirdly, regarding the number of light sensor channels in Table I, [29] utilised a light sensor with 256 channels. In contrast [30], [31], [32], and this work used 14, 6, and 10 channels respectively. Our headline error rate is approximately doubled compared to the very high channel count approach in [29]. This error comes at the benefit of more than three times the battery life while taking twice as many samples per unit time, highly desirable for long term experiments. Finally, all of the devices in Table I are below the 50 g weight value used as the lower limit for testing wearability comfort in [41].

Combining these findings from Table I, clearly design and experimental trade-offs are possible. This includes with cost, with our approach based strongly upon minimising cost. Different future works may prefer different operating points in the cost-accuracy-battery life trade-off space depending on the needs of the experiment being performed. Our approach to these trade-offs offers clear benefits for long term wrist based sensing. Measured results for 13 devices show a mean absolute error of less than 0.07 log lx, and a minimum between device correlation of 0.99.

This paper has focused on the system design and electronic performance testing of the new IoT sensor device. Its utility is evidenced by its use in [37] as a large, multi-user study. [37] recruited 59 participants and was able to demonstrate, amoungst other things, that higher pre-bedtime light exposure is associated with longer sleep onset latency, and early sleep timing is correlated with more reproducible and robust daily patterns of light exposure and higher daytime/lower night-time light exposure.

VI. Conclusion

This paper has presented a wearable light sensor node that can be incorporated into the IoT to provide monitoring of equivalent daylight illuminance exposure in real-world settings. It is optimized for wearing at the wrist to match standard smartwatch locations, and we have quantified the differences obtained to devices placed on other parts of the body. These findings demonstrate that accurate wrist worn light sensing, based upon CIE S026:2018, is feasible. This will allow mass data collection experiments, with a low cost point allowing a large number of participants. It can also allow personalised light exposure measurements to feed in to Internet-of-Things deployments for closed loop control of lightscapes in homes, offices, and other environments.

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